

Water and Wastewater Management at Indian Station Maitri in Antarctica

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Abstract

In accordance to environmental protocol to Antarctic treaty, monitoring of surface waters and wastewaters around Indian Antarctic station Maitri was carried out as a part of EI A studies during austral summer of 1993-94. Surface water sources like Zub lake, its feeding glacier waters, control lake and a seepage channel meeting to Zub lake were identified and samples were analysed for physical, inorganics (chloride, sulphate), nutrients (nitrate, phosphate), organics (TOC, hydrocarbon) and selective heavy metal levels. The results indicate that barring seepage water, all the samples possessed characteristics for drinking water qualities stipulated by WHO. However, glacier water at western side of the station showed an increasing trend of nutrients, dissolved solids, EC and heavy metals as compared to those to eastern side of the station. The above parameters increased substantially in seepage water indicating due attention to this channel of unknown origin.

Wastewaters generated by the station are treated aerobically in two biodiscs at enhanced temperature,. However, efficiency of treatment was recorded to be poor showing BOD reduction less than 50%. It has been estimated that 86% of the intake of Zub lake water is discharged as wastewater from biodiscs. Details of waste characteristics probable source of seepage water channel and possible impact of pumping of wastewater in land have been discussed.

Introduction

The Indian permanent station 'Maitri' is located at latitude 70°45'53"S & longitude 11 °44'3"E and situated on Schirmacher hills of East Antarctica. The station has been functioning smoothly since its commissioning in the austral summer of 1988-89 and has withstood the test of the time. While establishing this station, various environmental aspects had been taken into consideration to ensure that as little a damage to the environment is done.

There are innumerable number of freshwater lakes in the Antarctic continent. The 'Zub lake', formally known as 'Priyadarshini lake', covering an area

of approximately 0.35 sq km and located close to N-E side of Maitri station has been taken into consideration for its physico-chemical characteristics as it is the only source of drinking water to the inmates of Maitri station. The locations of summer camp urinal and bath room near the lake and a seepage channel emerging at higher elevation close to wastewater ponds & flowing towards the lake crossing the road may affect the drinking water resource. Besides, food packets, fuel drum, empty cans and other solid wastes were often located at the lake (Plate I). In view of this monitoring of this lake for its water quality has become the essential feature. Amongst various lakes located at Schirmacher Oasis, the one 1.5 km away towards N-W side of Maitri station (CLI) was selected as a control lake for assessing its original water quality. Similarly the feeder waters to the lake originated from glacier water (GL/L and GL/R) (Fig. 1) were also assessed.



Plate I: Fuel drums and food packets floating at Zub lake

Material and Methods

Site selection

Water : Four sampling locations at Zub lake were selected (Table 1) at north west, east and south directions, 100 m away from the bank of each side. One sampling location each from control lake and feeding waters to Zuh lake designated as GL/R ant! GL/L were also selected for water characterisation.

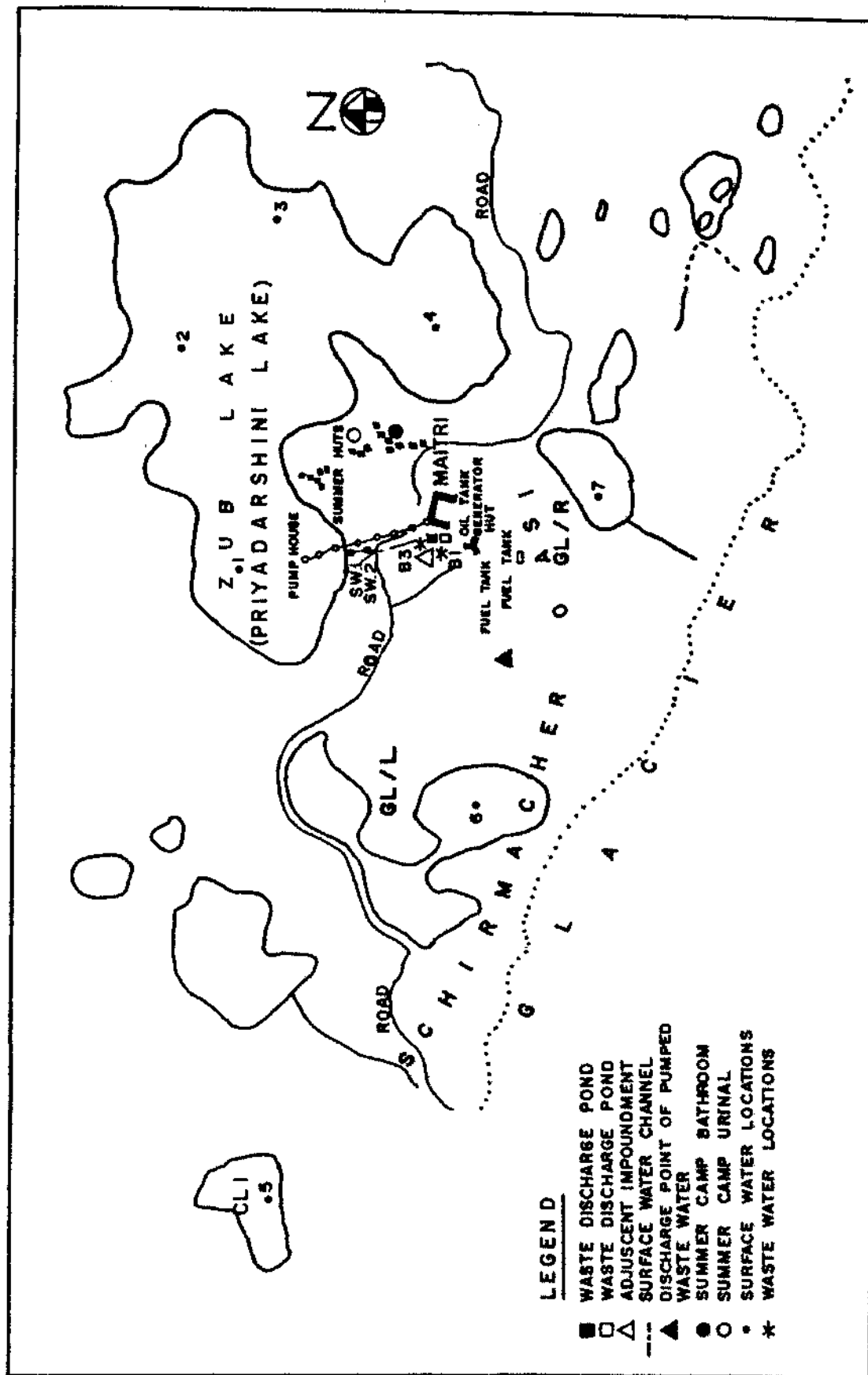


Fig. 1; Locations selected for water and waste water quality monitoring

Table 1: Surface Water Sampling Locations

S. No. Sampling location
Zub (Priyadarshini) Lake
1. Western side of the lake (100 m from the bank)
2. Northern side of the lake (100 m from the bank)
3. Eastern side of the lake (100 m from the bank)
4. Southern side of the lake (100 m from the bank)
Control Lake
5. Control lake No. 1 {1.5 km towards north-west side of Maitri station)
Glacier Lake
6. Southern side of glacier lake towards west of Maitri station
7. Southern side of glacier lake towards south-east of Maitri station
Seepage channel
8. Seepage channel emerging from impoundment (Sur)
9. Seepage channel emerging from unidentified source/through impoundment (SW2)

Besides, samples from seepage water channel joining at Zub lake were also collected and analysed.

Wastewater : The Zub lake water received by the Maitri station (2.90 to 4.96 m³/day) is used for different purposes and subsequently a sizable quantity is released as wastewater or grey water for further disposal in land through treatment in two biodiscs denoted as B1 and B3. The biodiscs are installed separately in two wooden rooms each measuring 6.5 m x 2.4 m x 3.25 m (h) for protection against snow deposition. While B1, which was installed in 1988 caters for the kitchen wastes, B3 installed in 1989 receives wastes from bathroom, urinal, wash basins and washing machines. In general, 250 kg detergent and 50 kg vim powder are used in a year by the inmates of the station. While the overall size of B1 treatment plant is 1.6 m x 1.75 m, its inside chamber measures as 1.25 m x 1 m x 1.2 m deep. It consists of 47 square discs each of 490 cm² and three circular discs (each of 95 cm diameter) made of GRP sheets (Plate 2). The size of B3 is comparatively larger, the chamber being 1.5 m x 1.75 m x 1.2 m deep with an overall size 2.1 x 2.78 m. It carries 71 square (1 m² each) and three circular (each 1.4 m diameter) discs. In order to increase the efficiency of treatment aerobically, provision has been made in both the biodiscs to enhance the temperature of the reactors at about 24°C by inserting 2000 W immersion heaters. The treated wastewaters are collected at two ponds designated as B1 (19 m x 20 m x 2 m) and B3 (12.4 m x 20 m x 2 m) representatives of respective biodiscs.



Plate 2: A portion of Biodise (B1) showing inner chamber and type of discs

Another impoundment (13 m x 13 m) which was earlier used as waste collection pond at N-W side of B3 pond is abandoned since 1993. The collected wastes at B1 and B3 ponds are pumped out on every 8 to 12 days on barren land at a distance of 190 m away towards S-W side of the station. It may be mentioned that the distance between this pumping site and nearest glacier lake which feeds to Zub lake is 210 m. Considering the level of Zub lake water as base, the elevations of B1/B3 ponds, pumping site and nearest glacier lake water level stand as 13, 16 and 4 m respectively. The distance between waste collecting ponds and nearest Zub lake water is 150 m. In view of lower elevation, a surface seepage channel was observed to be occurring to flow towards Zub lake (Plate 3). Flow rate of this seepage which supports thick algal growth becomes maximum during the months November to January, then gradually reduces and finally freezes in the middle of February. On February 5, 1994, the flow rate was measured as 40 l per hour.

Besides, wastewaters were also generated from kitchen, urinal and bathroom of summer hut in 1993-94. Since the quantity was not enough to treat, these were allowed to seep underground.

Sample collection and analysis

Water : On each sampling occasion, surface water samples were collected by means of a clean polyethylene bucket (20 litre capacity) during Jan.-Feb. 1994



Plate 3: The seepage water channel crosses the road and finally joins the Zub lake

and transferred into different clean polyethylene bottles as per the parameters selected for water quality. Field parameters like pH, temperature, conductivity, turbidity and dissolved oxygen were estimated immediately after collecting the samples by means of century portable kit. Samples collected for other parameters like phosphate, nitrate, nitrite, ammonia, total organic carbon and heavy metals were preserved as per standard methods (1992). Separate samples were collected for the estimation of hydrocarbon and extracted with carbon tetrachloride. The extracts were preserved and brought back to Nagpur for estimation through fluorescence spectrophotometer (HITACHI, F-4000).

Wastewater : The water intake from Zub lake to Maitri station for designated purposes was found to be 2.90 to 4.96 m³/day. The quantity of effluents generated by two biodiscs in February 1994 was measured at different time intervals and variations between zero to zero discharge are depicted in Fig.2. It has been estimated that 86% of intake water from Zub lake is discharged as wastewater through the biodiscs. The proportion appears to be marginally high at Antarctica as compared to that in India. The pumping rate of the wastewater which was accumulated for 10 days in ponds during early February 1994 and was discharged on land subsequently, was measured as 28.8 m³, i.e. 2.88 m³ day⁻¹. While comparing this value with the effluent discharged by biodiscs, it appears that 13.9% of wastewater was lost due to seepage through 2 waste collecting ponds. Measurement of flow in seepage water channel (around 1 m³/day) during this period indicated that the rate of seepage from waste

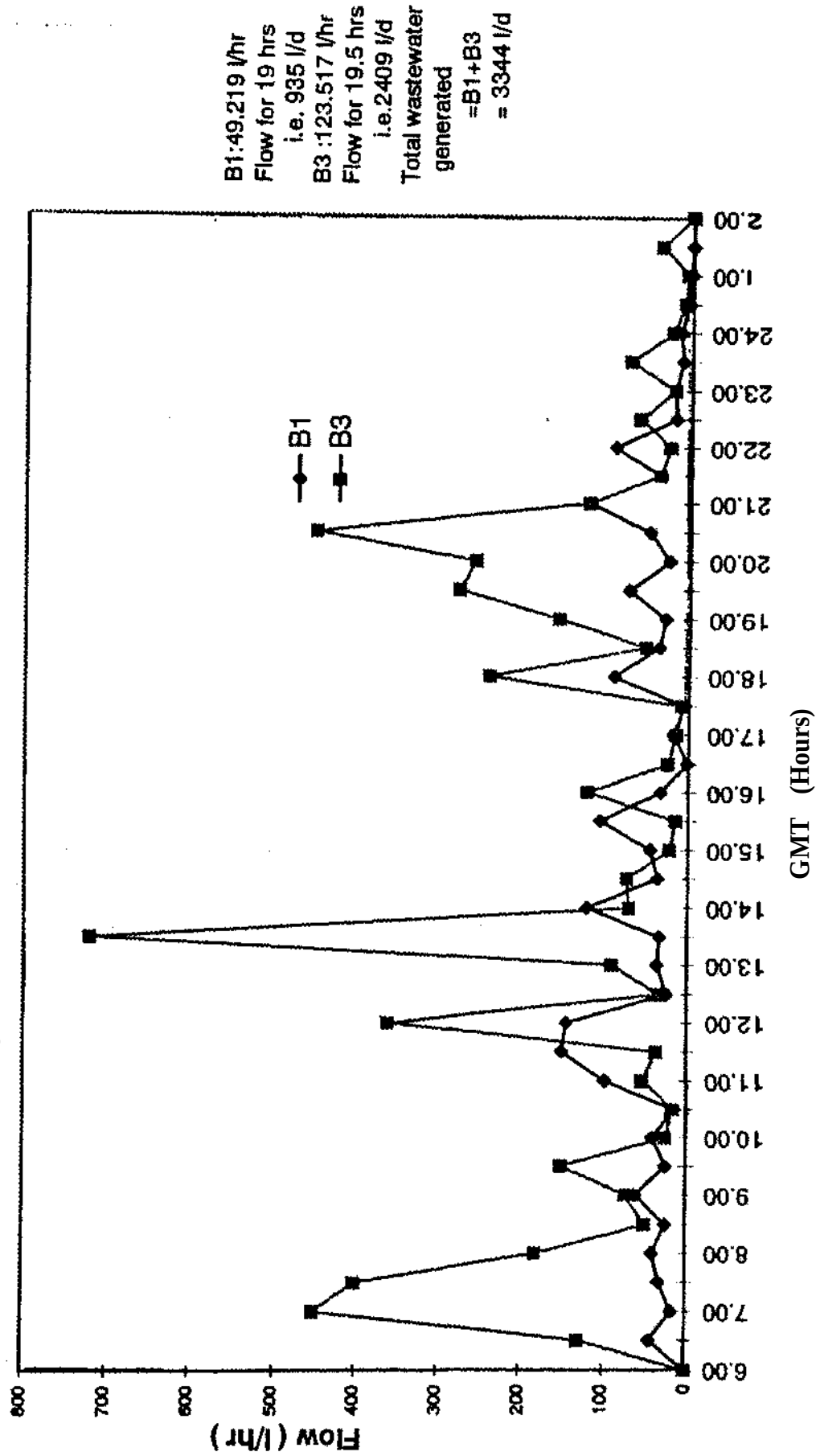


Fig.2: Flow of waste water through Biodiscs B1 & B3 installed in Maitri at different time in February, 1994

collecting ponds (B1 & B3) was almost half of the flow of seepage water channel leading to Zub lake. Sufficient data on these aspects need to be collected for assuring the source of seepage water channel.

The influent and effluent wastewater samples were collected from bio-discs B1 and B3 for specific parameters, viz. pH, total suspended solids, chloride, sulphate, nutrient load, heavy metal and chemical & biochemical oxygen demand.

Results and Discussions

Water : The physico-chemical characteristics of the water samples from Priyadarshini lake, control lake, glacier lakes and seepage water channel are shown in Table 2. The physical characteristic in terms of pH indicates acidic *pH ranging from 4.6 to 6.7. The acidic nature of waters from glacier lakes* indicates the characteristics of precipitated water with low conductivity, turbidity and dissolved solids whereas slight increase in pH of Zub lake indicates the dissolution of minerals through inlet run off waters and thereby increase in conductivity and dissolved solids.

The inorganic parameters in terms of sulphate and chloride were found in the range of 10-29 mg/l whereas nutrients in terms of phosphate and nitrate were recorded in the range of ND-0.80 µg/l and 0.01 to 0.08 mg/l respectively. Dissolved oxygen levels varying from 8.2 to 8.7 mg/l indicate better reaeration capacity with negligible photosynthetic activities in the lake. Though concentrations of chloride, nitrate and total organic carbon in the Zub lake water were found to be well within permissible limits, the values warrant the trend of enhancement as compared to control and glacier lakes probably because of incoming seepage and decomposition of moss at surrounding areas. It has further been noticed that dissolved solids, EC and nutrient concentrations at glacier lake water (western side) was comparatively higher from those of similar type of water located at eastern side of the station (Table 2). Since the former water is at a distance of 210 m from the location where wastewater at a rate of about 3 m day⁻¹ is pumped out, there is a possibility of contamination of glacier water by seeped wastewater.

Heavy metal concentrations in terms of cadmium, copper, chromium, lead and zinc were also recorded within permissible limits for drinking water.

The physico-chemical characteristics of seepage channel waters (SW1 & SW2) indicate that the water possesses comparatively higher levels of dissolved solids (150 and 1550 mg/l), conductivity, nutrients, suspended solids as also marginal increase of organic load in terms of organic carbon & hydrocarbons (Table 2). Heavy metal concentrations were also found to be present in both

the samples indicating its probable origin from wastewater impoundments which warrants to pollute the lake in future.

Wastewater : The physico-chemical characteristics of wastewater samples from influent and effluent of biodiscs B1 and B3 are shown in Table 3. The data indicate that the influents of both the biodiscs show alkaline *pH* with high BOD (490 and 540 mg/l) & COD (1510 and 1650 mg/l) and marginal nutrient load. However, the effluents coming out from both the discs show acidic *pH* (5.5 and 6.5). This might be attributed to degradation of certain

Table 2: Abiotic Characteristics of Water Samples from Zub Lake, Control Lake, Glacier Lakes and Seepage Channel

Sr. No.	Parameter	Unit	Concentration levels								
			Zub lake				Control lake		Glacier lake	Seepage channel	
			1	2	3	4	5	6	7	SW ₁	SW ₂
A.	Physical										
	Turbidity	NTU	3.4	3.3	3.0	2.4	4.6	4.6	4.6	65.7	34.8
	pH	—	5.8	5.7	5.1	6.0	6.7	4.6	4.6	6.55	6.22
	Conductivity	us/cm	21.3	18.6	24.04	20.06	10.6	9.3	8.2	105.0	1377.0
	Total dissolved Solids	mg/l	12.9	11.2	10.3	9.4	3.0	10.0	6.0	150.0	1550.0
	Suspended solids	mg/l	4.0	2.0	5.0	3.0	2.0	10.0	6.0	40.0	74.0
B.	Inorganic										
	Chloride	mg/l	23.0	24.0	29.0	23.0	11.0	11.0	10.0	20.3	3.8
	Sulphate	mg/l	4.0	1.0	4.0	4.0	9.0	9.0	8.0	9.7	15.0
C.	Nutrient & Demand										
	Nitrate	mg/l	0.07	0.01	0.02	0.06	0.05	0.08	0.05	0.22	0.46
	• Nitrite	mg/l	ND	ND	ND	ND	ND	ND	ND	ND	ND
	Phosphate	mg/l	ND	ND	0.08	ND	ND	0.1	ND	0.09	0.72
	Dissolved oxygen	mg/l	8.5	8.2	8.6	8.4	8.7	8.4	8.4	8.2	8.1
D.	Organic										
	Total organic carbon	mg/l	10.0	15.0	10.0	8.5	ND	ND	ND	11.5	1205
	Hydrocarbon	µg/l	0.03	0.03	0.03	0.03	0.01	0.01	0.01	0.25	0.07
E.	Heavy metals										
	Cadmium	µg/l	7.0	15.0	6.0	13.0	6.0	3.0	3.0	0.10	ND
	Copper	µg/l	ND	2.0	2.0	32.0	4.0	9.0	3.0	1.01	0.06
	Chromium	µg/l	14.0	22.0	ND	4.0	ND	14.0	ND	ND	0.30
	Lead	µg/l	8.0	27.0	ND	58.0	53.0	64.0	58.0	ND	0.01
	Iron	µg/l	256.0	251.0	140.0	343.0	200.0	70.0	98.0	4.33	3.13
	Manganese	µg/l	2.0	41.0	ND	199.0	ND	43.0	22.0	0.16	1.48
	Zinc	µg/l	77.0	113.0	39.0	99.0	47.0	129.0	102.0	0.04	0.92

SW₁: Seepage channel confluence to Zub lake; SW₂: Seepage channel 2m before confluence; ND: Not detectable

**Table 3 : Physico-chemical Characteristics of Wastewater Sample
(except pH, all values are expressed in mg/1)**

Sr.NO.	Parameter	Biodisc (B1)		Biodisc (B3)	
		Influent	Effluent	Influent	Effluent
1.	pH	7.98	5.50	8.11	6.5
2.	Suspended solids	646.06	124.00	68.00	30.00
3.	Chloride	63.00	52.00	110.00	85.00
4.	Sulphate	18.5	10.6	55.05	97.32
5.	Phosphate	0.52	0.73	4.2	5.2
6.	Nitrate	8.3	6.4	24.3	16.8
7.	Chemical oxygen demand	1510.0	920.0	1650.0	1100.0
8.	Biochemical oxygen demand	490.0	310.0	540.0	400.0

Table 4 : Levels of Heavy Metals in Wastewaters of Biodiscs

Sr. NO.	Sampling location	Waste	Cadmium	Conner	Chromium (mg/1)	Lead	Iron	Zinc
1.	Biodisc (B1) sample collected at time t ₂	I	0.01	0.06	0.24	0.26	16.74	2.31
		E	ND	1.52	0.27	0.02	21.46	11.26
2.	Biodisc (B1) sample collected at time t ₂	I	ND	0.03	0.12	0.01	13.57	9.84
		E	ND	0.05	0.12	0.02	11.25	6.32
3.	Biodisc (B1) composite sample	I	ND	0.33	0.11	0.09	4.02	10.36
		E	ND	0.65	0.14	0.01	6.25	8.35
4.	Bicxlisc (B3) sample collected at time t ₁	1	ND	0.18	0.27	ND	3.34	1.05
		E	ND	0.28	0.06	0.09	4.02	0.01
5.	Biodisc (B3) sample collected at time t ₂	I	ND	0.84	0.30	0.03	4.53	14.72
		E	0.01	0.06	0.24	0.26	6.11	0.35
6.	Biodisc (B3) composite sample	I	ND	0.84	0.19	0.09	4.01	10.36
		E	ND	0.16	0.12	0.09	4.16	0.10

I: influent; E; Effluent; ND Not Detectable

sulphurous and nitrogenous compounds converting into gaseous forms and altering the water in acidic range. The levels of organics in the effluents did not reduce remarkably in the plant, showing BOD 310 & 400 mg/1 and COD 920 & 1100 mg/1 respectively for B1 and B3 biodiscs. The marginal reduction (less

than 50%) in BOD and COD of the influents indicates poor efficiency of the biodiscs which needs detailed investigations during future expeditions. Often the discs were observed not in rotating state.

The data on heavy metals from wastewaters of biodiscs B1 and B3 are shown in Table 4. The heavy metal concentrations of influents of both the discs were found in the range of ND-0.01 for cadmium, 0.03-0.84 for copper, 0.11 -0.30 for chromium, ND- 0.26 for lead, 3.34-16.74 for iron and 1.05-14.72 mg/l for zinc. However, the heavy metal concentrations of effluent samples were measured in the range of ND-0.01 for cadmium, 0.05-0.65 for copper, 0.06-0.24 for chromium, 0.01 -0.26 for lead, 4.02-11.25 for iron and 0.01 -11.26 mg/l for zinc. The high concentrations of iron, zinc, lead, copper and chromium in wastewaters from both the biodiscs might be due to the materials regularly used in the station as also the dissolution of the corroded materials either from pumping pipes or biodiscs.

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